



Universidade do Minho
Departamento de Informática

Redes Neurais Artificiais @ KNIME

with Keras

LEI @ 2025/2026, 2º sem



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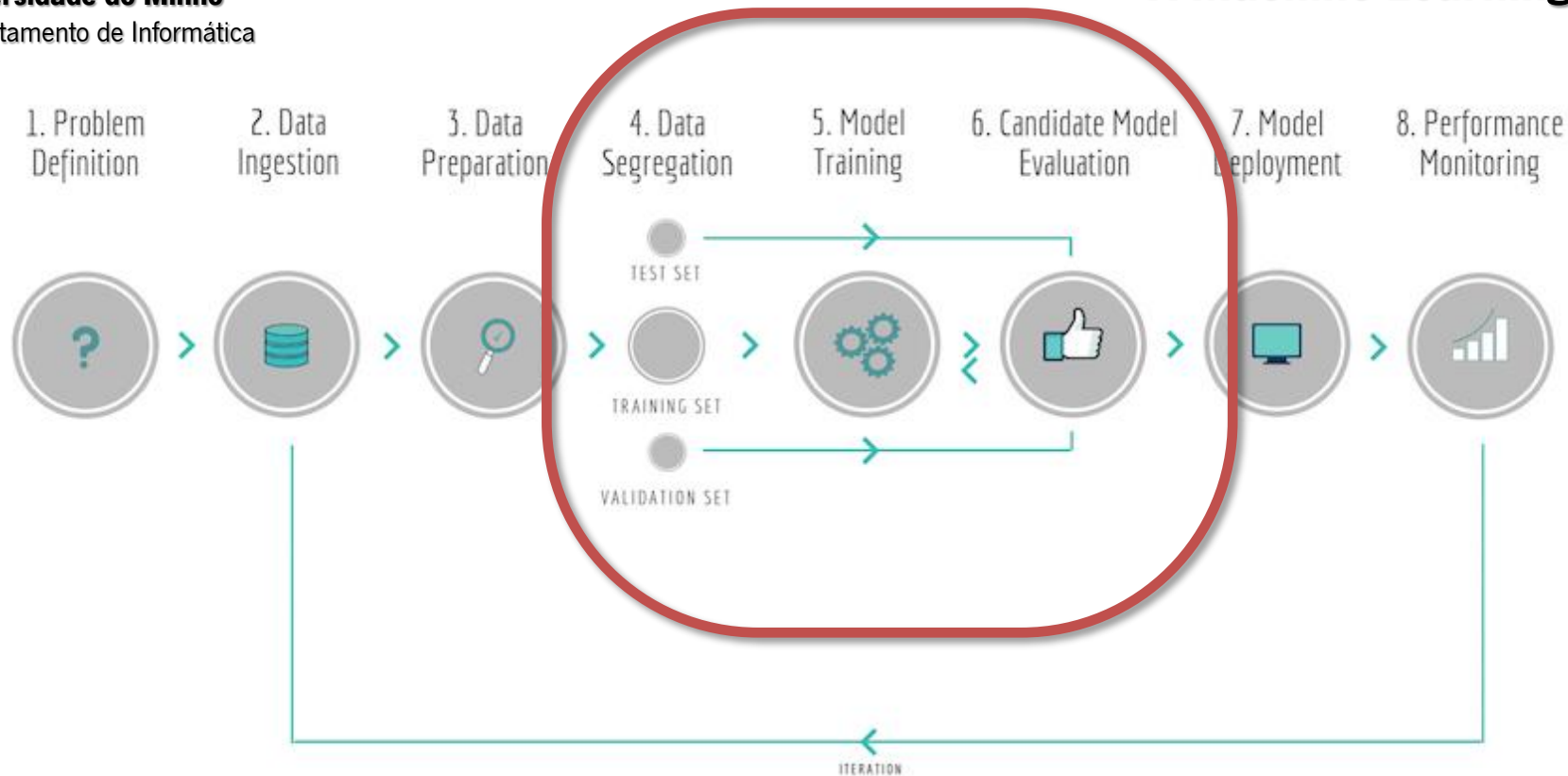
Agenda

- O fluxo Learner-Predictor para Redes Neurais em KNIME
- MLP: Multi-layer perceptron em KNIME
 - KNIME Keras Integration
(File/Install Knime Extensions...)
- Experimentação
(*hands on*)





A Machine Learning Pipeline



(<https://towardsdatascience.com/architecting-a-machine-learning-pipeline-a847f094d1c7>)



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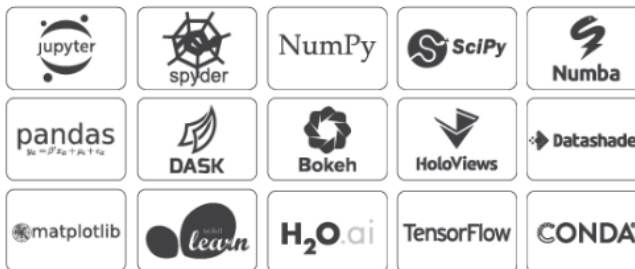
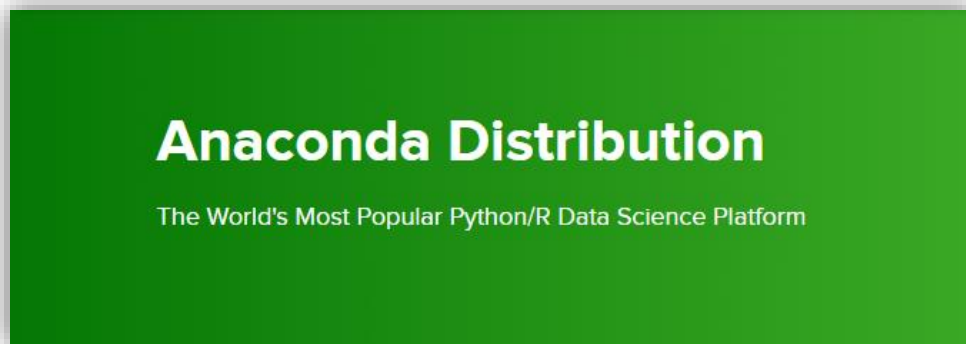
- A extensão KNIME “Deep Learning – Keras Integration” permite maior controlo sobre os parâmetros de criação de uma RNA:
 - Topologia
 - Estratégia de inicialização dos pesos (valores das sinapses)
 - Regras de aprendizagem
 - Iterações de treino
 - Taxa de aprendizagem
 - ... e muitos outros parâmetros!



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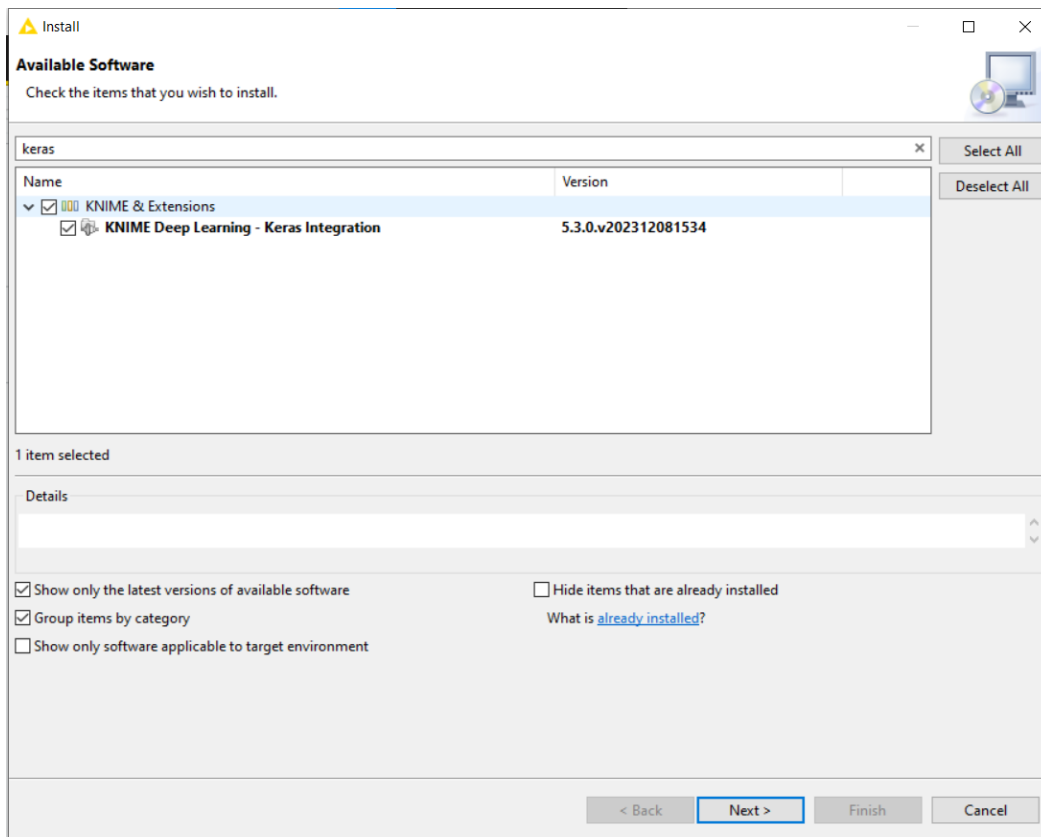
- Instalar o ambiente Anaconda: [Anaconda Download](#)





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Instalar **KNIME Deep Learning – Keras Integration** extension



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■ Criar o ambiente Anaconda

The image shows the 'Python Deep Learning' preferences window in KNIME. On the left, a tree view under 'KNIME' has 'Python Deep Learning' selected, indicated by a red arrow from a blue box labeled 'Selecionar a opção'. The main panel shows options for using a special Deep Learning configuration. Under 'Deep Learning Python environment configuration', 'Conda' is selected. For both 'Keras' and 'TensorFlow 2', the 'Name of the ... Conda environment' is set to 'py3_knime_dl', and a red arrow from a blue box labeled 'Criar o ambiente' points to the 'New environment...' button. Another red arrow from a blue box labeled 'Selecionar o ambiente criado' points to the 'py3_knime_dl' dropdown menu.

Selecionar a opção

Criar o ambiente

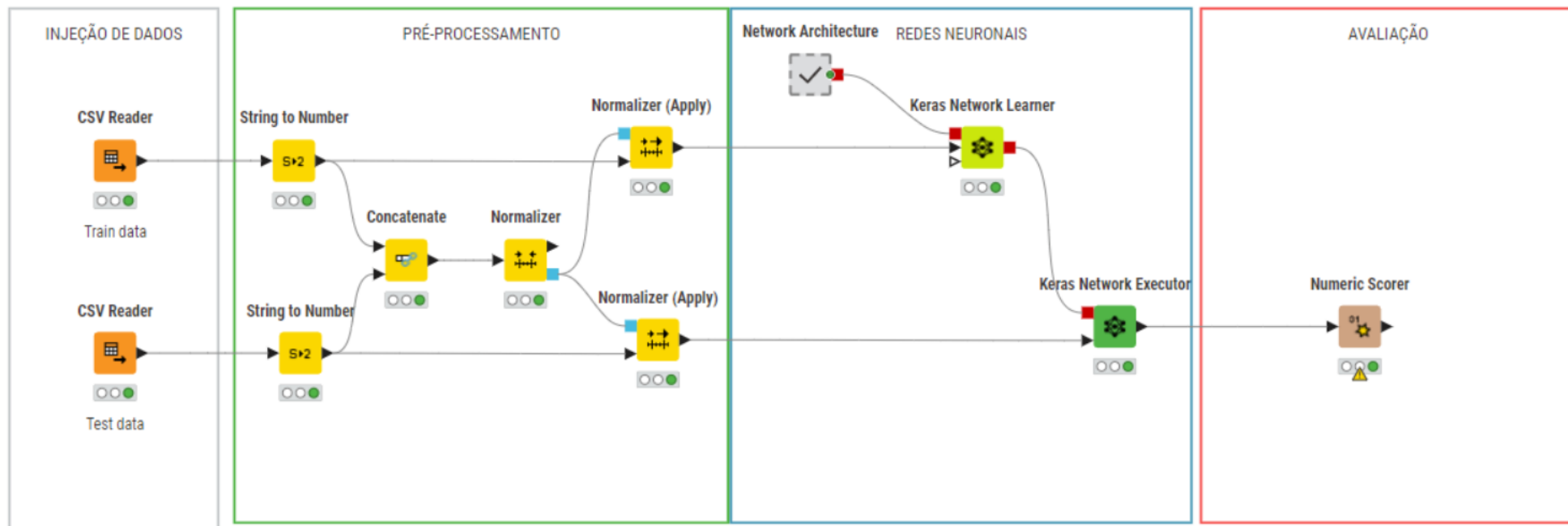
Selecionar o ambiente criado



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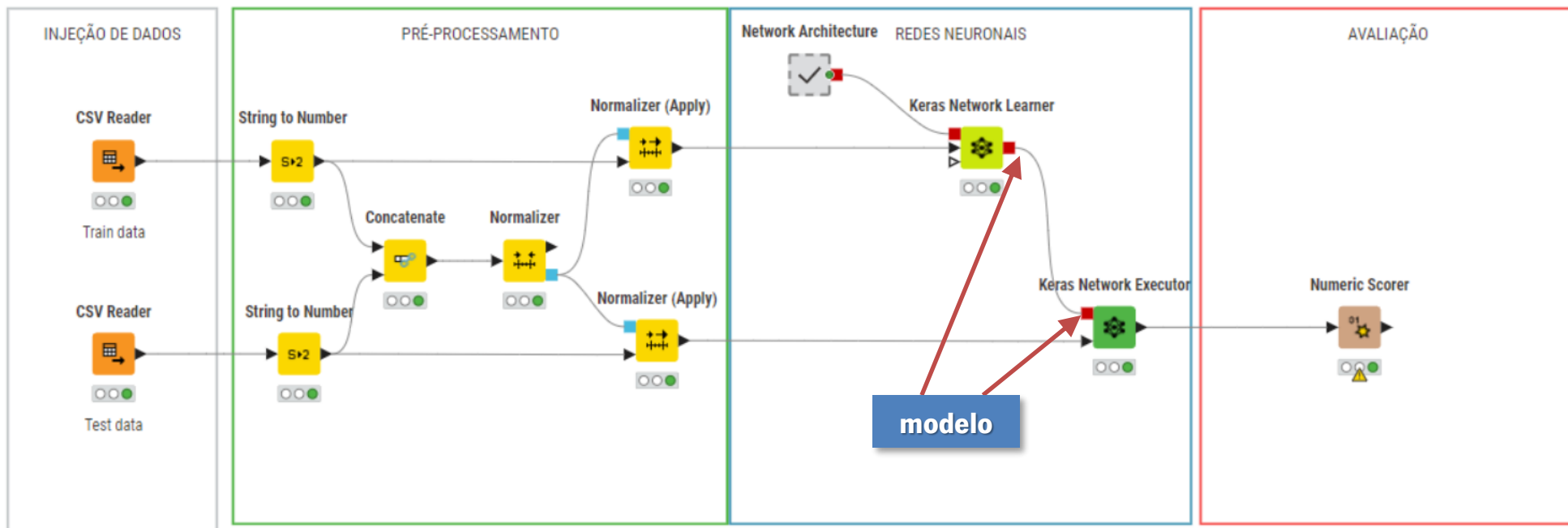
(Keras Integration)

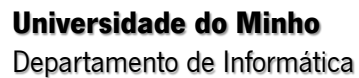




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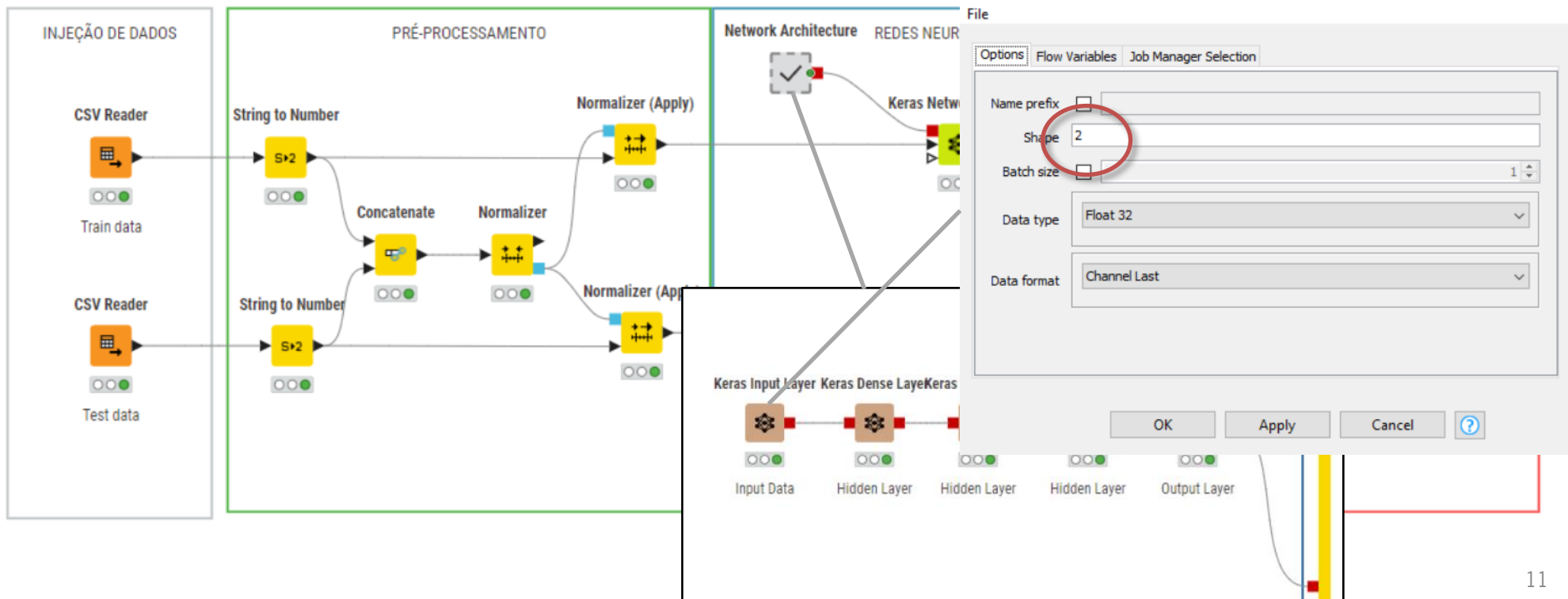
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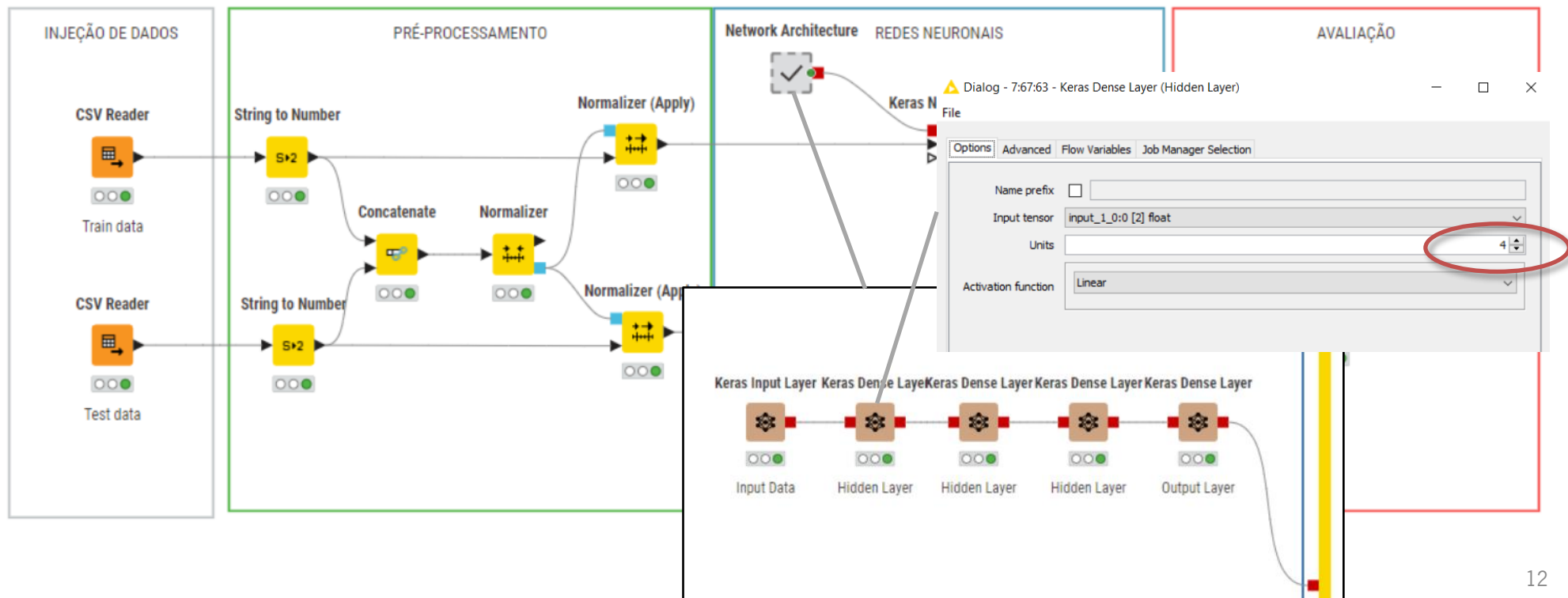
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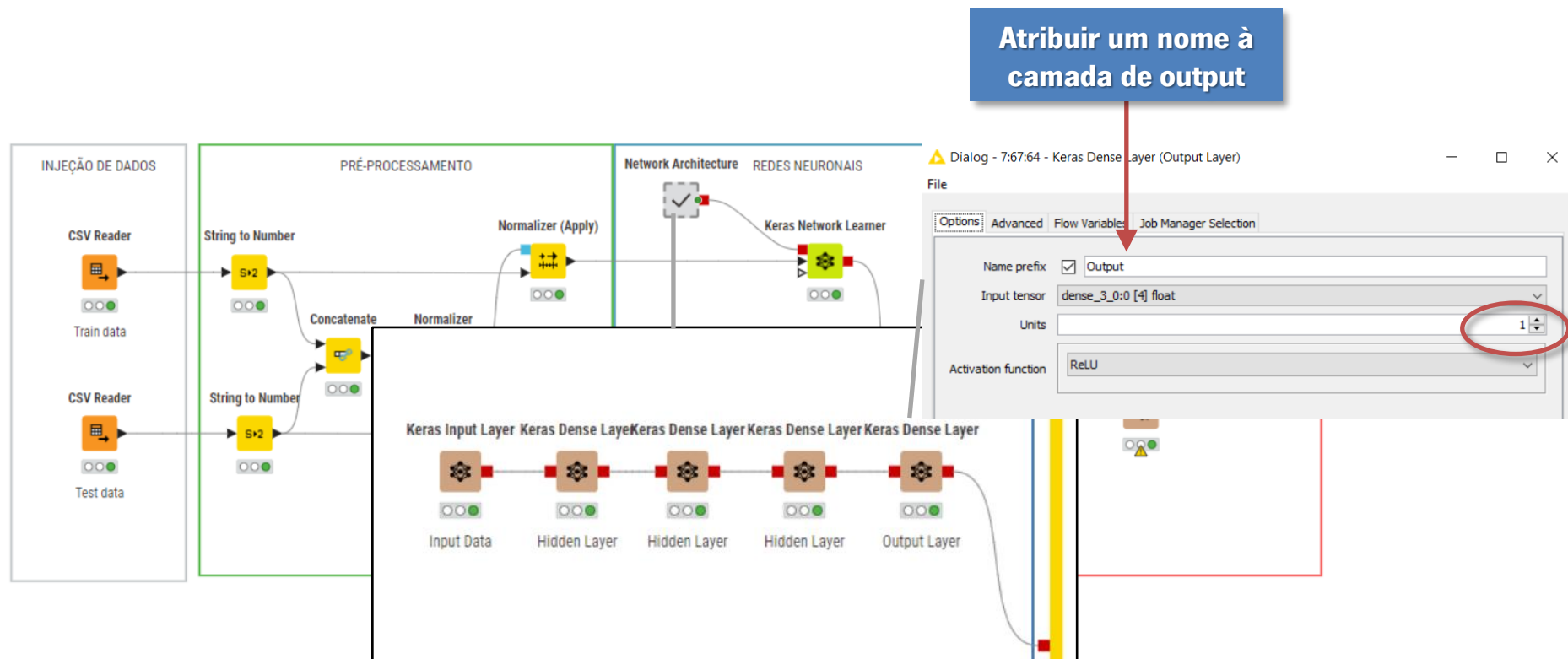
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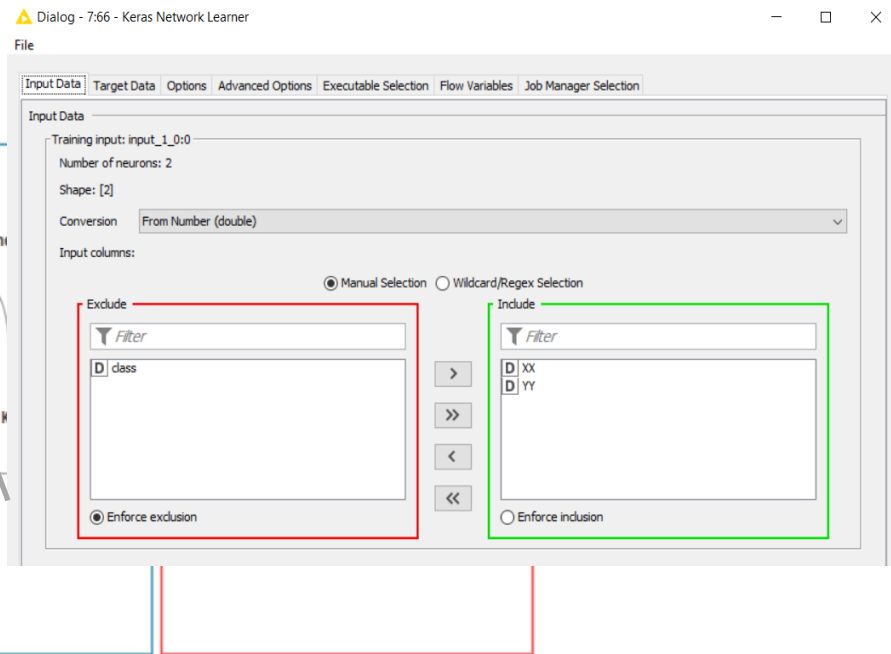
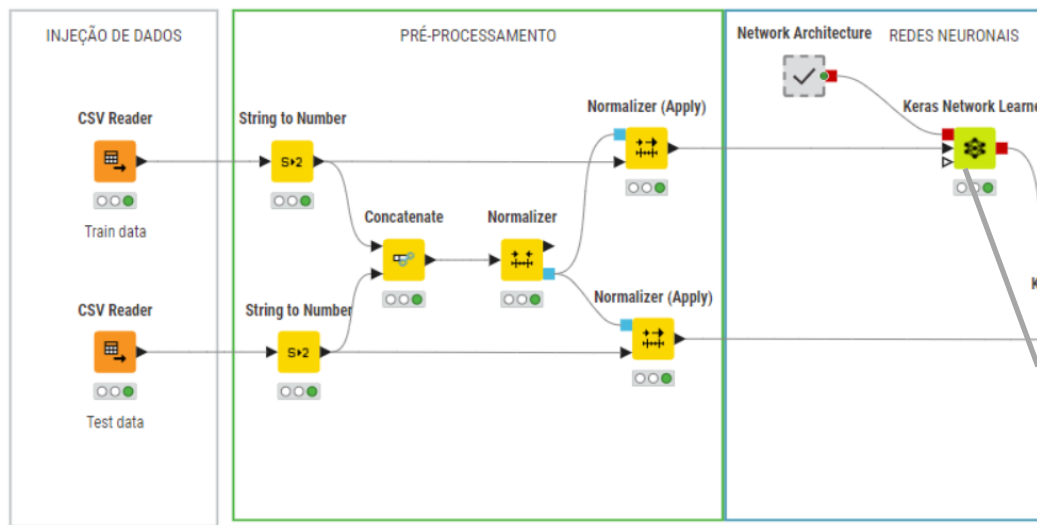




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(Keras Integration)

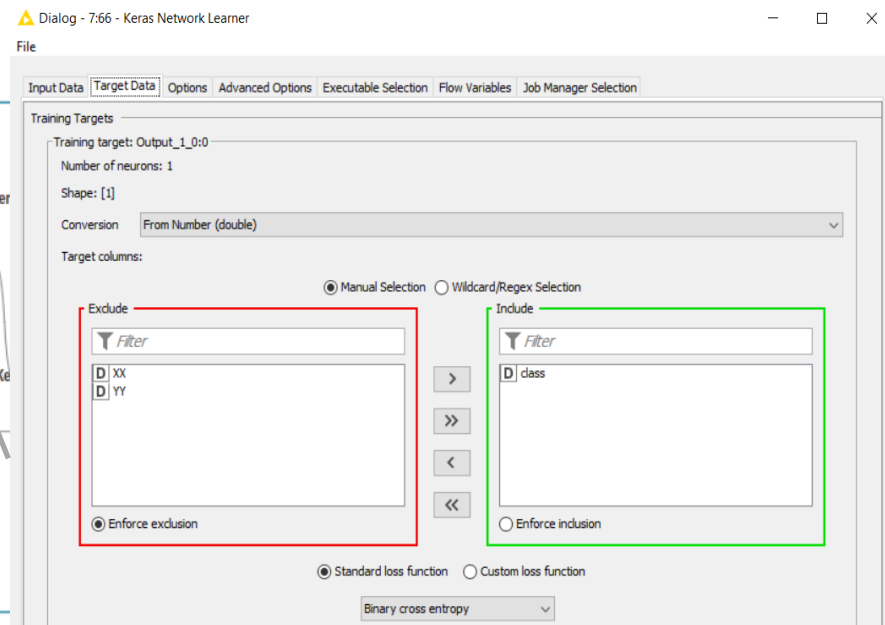
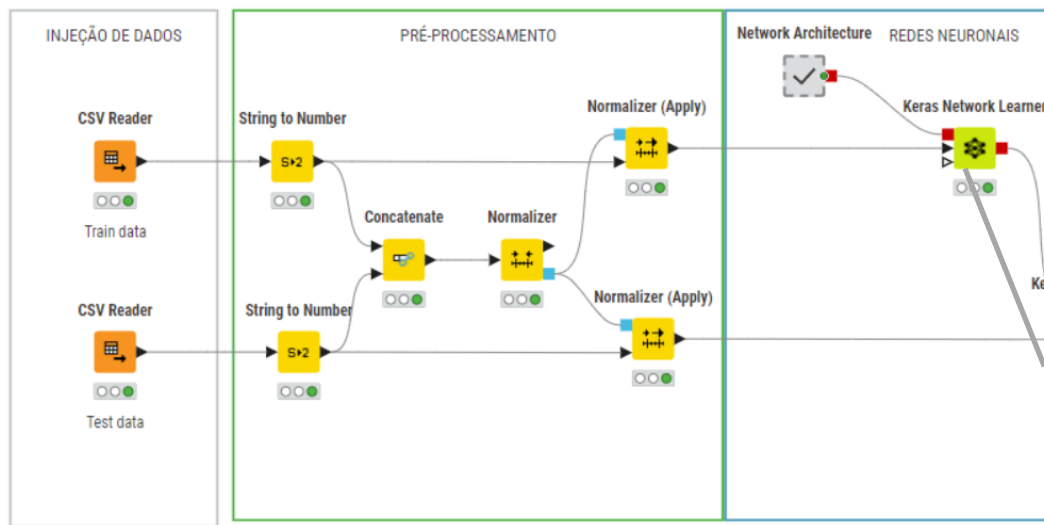




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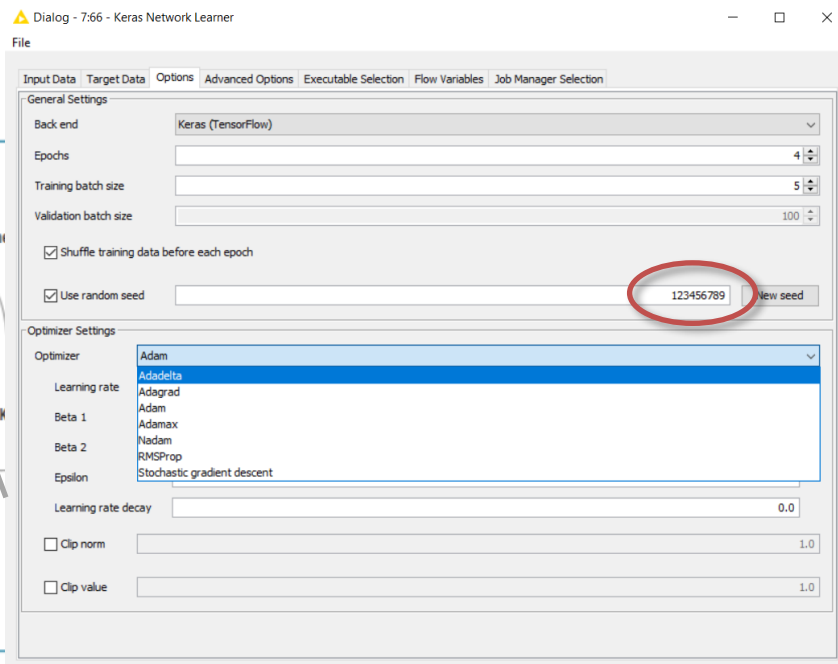
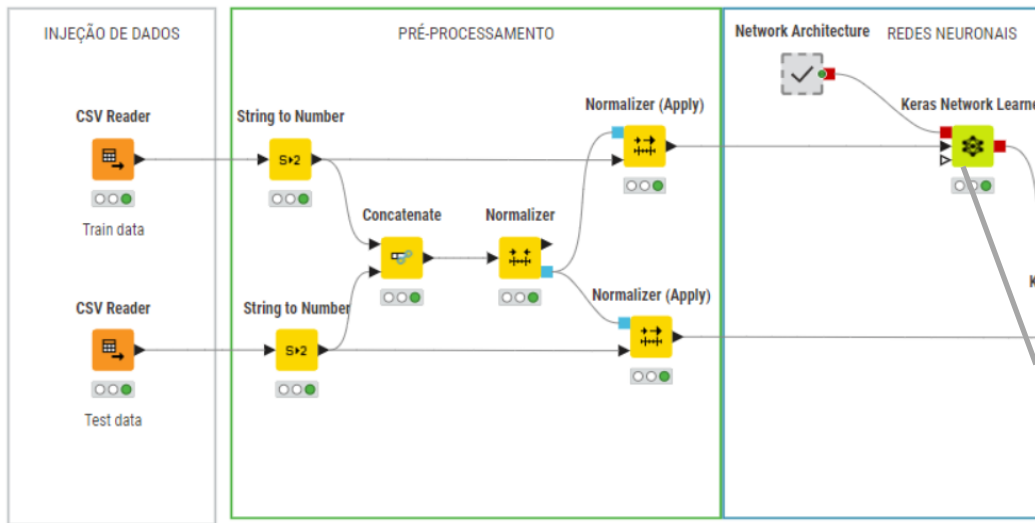
(Keras Integration)





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Dialog - 4:65 - Keras Network Executor

File

Options: Advanced Options Executable Selection Flow Variables Job Manager Selection Memory Policy

General Settings

Back end: Keras (TensorFlow)

Input batch size: 100

☒ Keep input columns in output table

Inputs

Input: input_1:0

Number of neurons: 2

Shape: [2]

Conversion: From Number (double)

Input columns:

☒ Manual Selection ☐ Wildcard/Regex Selection

Exclude

Filter

D class

☒ Enforce exclusion

Include

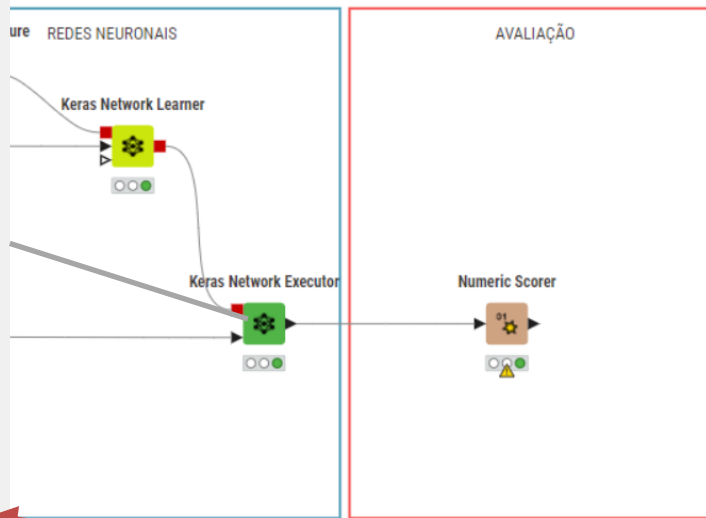
Filter

D XX
D YY

☐ Enforce inclusion

Outputs

add output

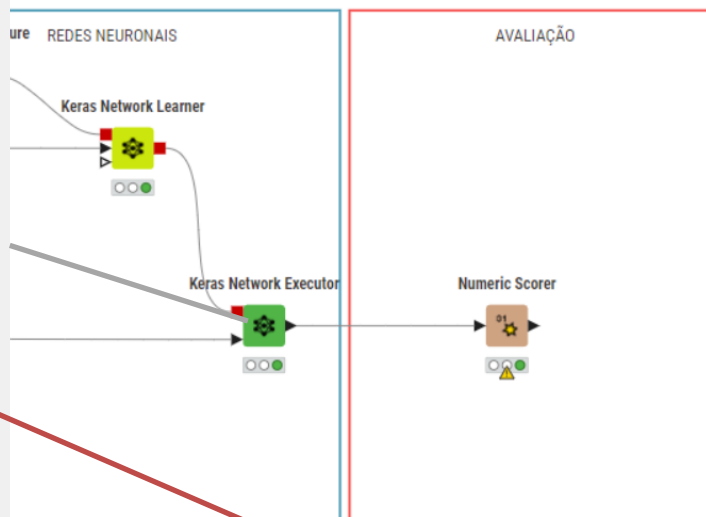
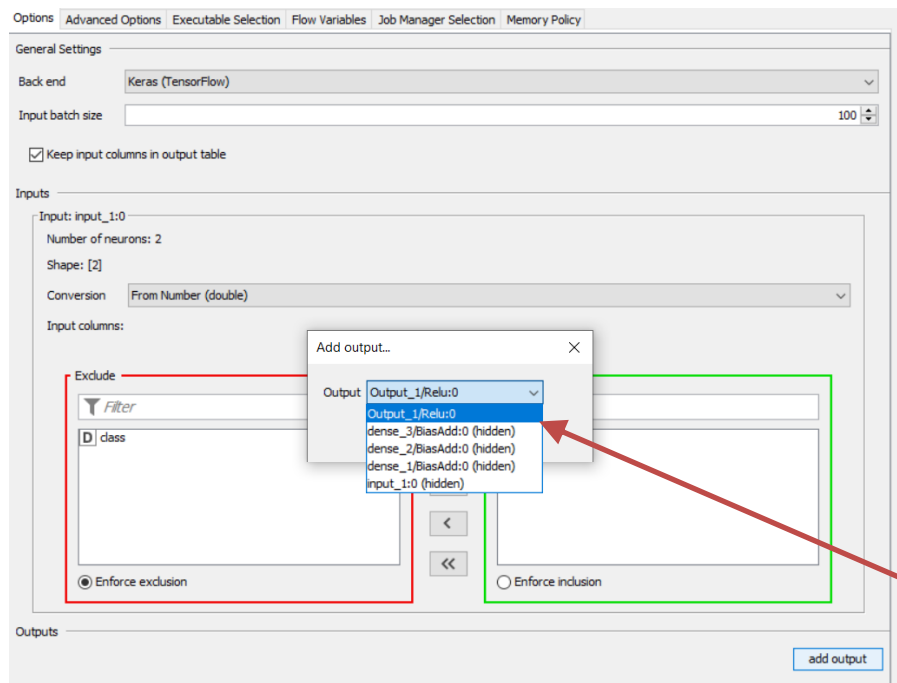


**Adicionar a camada
de output**



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**Selecionar a camada de output
renomeada anteriormente**



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Dialog - 7:65 - Keras Network Executor

File

Options | Advanced Options | Executable Selection | Flow Variables | Job Manager Selection | Memory Policy

Back end: Keras (TensorFlow)

Input batch size: 100

☒ Keep input columns in output table

Inputs

Input: input_1:0
Number of neurons: 2
Shape: [2]
Conversion: From Number (double)

Input columns:

☒ Manual Selection ☐ Wildcard/Regex Selection

Exclude

Filter

D | class

☒ Enforce exclusion

Include

Filter

D | xx
D | yy

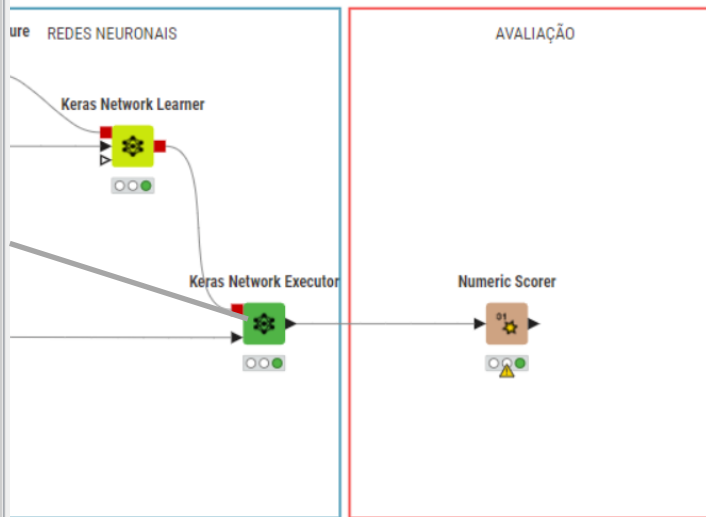
☐ Enforce inclusion

Outputs

add output

remove

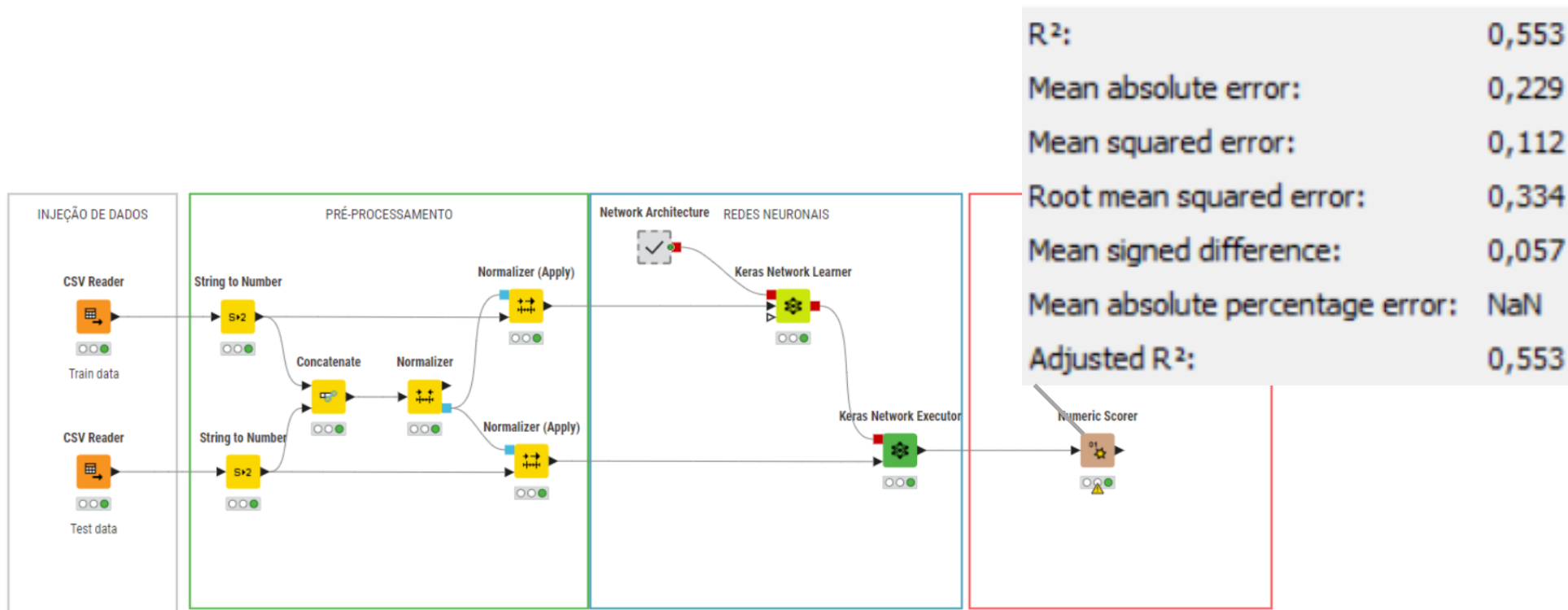
Output: Output_1/Relu:0
Shape: [1]
Conversion: To Number (double)
Output columns prefix: Output_1/Relu:0_





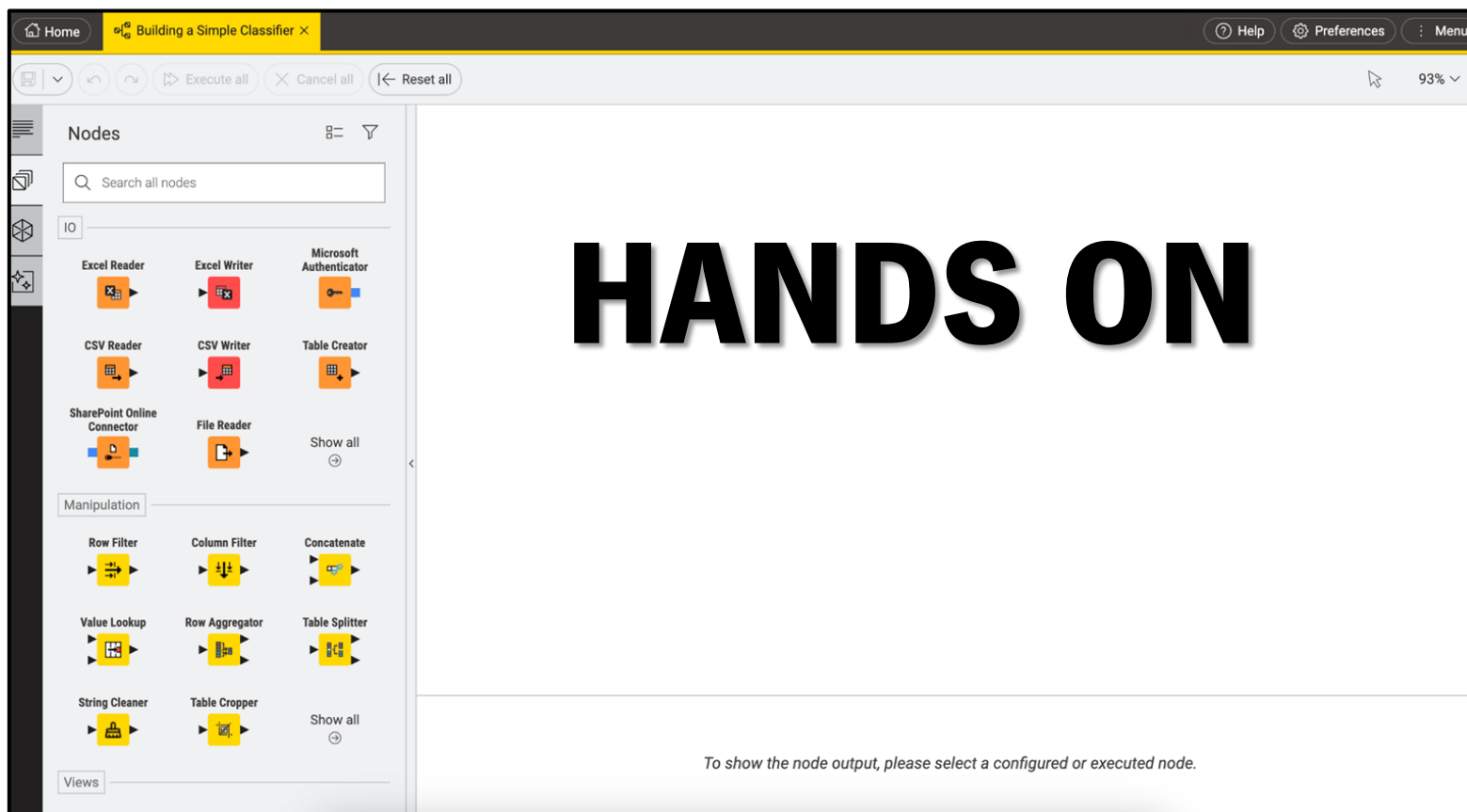
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